

A NOTE ON RETRACTS OF POLYNOMIAL RINGS IN THREE VARIABLES

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ABSTRACT. For retracts of the polynomial ring, in [1], Costa asks us whether every retract of $k^{[n]}$ is also the polynomial ring or not, where k is a field. In this paper, we give an affirmative answer in the case where k is a field of characteristic zero and $n = 3$.

1. INTRODUCTION

Let A and B be commutative rings. We say A is a *retract* of B if A is a subring of B and there exists an ideal I of B such that $B \cong A \oplus I$ as A -modules. The followings are basic properties of retracts.

Proposition 1.1. (cf. [1, Section 1]) *Let B be an integral domain and let A be a retract of B . Then the following assertions hold true.*

- (1) *A is algebraically closed in B .*
- (2) *If B is a UFD, then A is also a UFD.*
- (3) *If B is regular, then A is also regular.*

Lemma 1.2. *Let k be a field. Let A and B be k -algebras. If A is a retract of B , then $A \otimes_k K$ is a retract of $B \otimes_k K$ for any field K containing k .*

Proof. Since A is a retract of B , there exists an ideal I of B such that $B \cong A \oplus I$ as A -modules. Let K be a field containing k . Taking a tensor product by K over k , we have $B \otimes_k K \cong (A \otimes_k K) \oplus I'$ as $A \otimes_k K$ -modules, where $I' := I \otimes_k K$ is an ideal of $B \otimes_k K$. Thus, $A \otimes_k K$ is a retract of $B \otimes_k K$. $\square$

Let k be a field. We denote $k^{[n]}$ by the polynomial ring in n variables over k . In [1], Costa asks us the following question.

Question 1.3. *Let k be a field and let $B := k^{[n]}$. Then, is every retract of B containing k the polynomial ring?*

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If $n \leq 2$, then the above question is affirmative and proved by Costa ([1, Theorem 3.5]). The main result in this paper is to give an affirmative answer for Question 1.3 in the case where k is a field of characteristic zero and $n = 3$.

2. MAIN RESULTS

Let k be an algebraically closed field and let X be a (not necessarily complete) nonsingular algebraic variety over k . By virtue of Nagata's Completion Theorem ([8]), there exists a complete algebraic variety $\overline{X}$ over k such that X is open and dense in $\overline{X}$. We say that X is a *resolvable variety* if Hironaka's Main Theorems about resolution of singularities hold for $\overline{X}$ and $\overline{X} - X$. For a resolvable variety X , we denote $\bar{\kappa}(X)$ by the *logarithmic Kodaira dimension*.

Lemma 2.1. (cf. [5, Theorem 1.1 (a)]) *Let $f : X \rightarrow Y$ be a morphism of nonsingular, resolvable algebraic varieties over an algebraically closed field. If f is dominant and generically separable, then $\bar{\kappa}(X) \geq \bar{\kappa}(Y)$.*

The following is a characterization for affine planes (see Miyanishi [6], Fujita [3], Miyanishi–Sugie [7] and Russell [9]).

Theorem 2.2. *Let k be an algebraically closed field and X be a nonsingular affine surface over k . Let A be the coordinate ring of X , namely $X = \operatorname{Spec} A$. Then $X \cong \mathbb{A}_k^2$ if and only if $A^* = k^*$, A is a UFD and $\bar{\kappa}(X) = -\infty$.*

First of all, we shall show some properties of retracts of the polynomial ring over a field.

Lemma 2.3. *Let k be an algebraically closed field and let $B := k^{[n]}$ be the polynomial ring in n variables over k . Let A be a retract of B containing k and set $X = \operatorname{Spec} A$. If X is resolvable and $Q(B)$ is separably generated over $Q(A)$, then the following assertions hold true.*

- (1) *A is a finitely generated UFD over k with $A^* = k^*$,*
- (2) *X is a nonsingular variety over k ,*
- (3) *$\bar{\kappa}(X) = -\infty$,*

where we denote $Q(R)$ by the quotient field of an integral domain R .

Proof. Since A is a retract of B , A is a k -subalgebra of B . Hence it is clear that A is an integral domain with $A^* = k^*$. Furthermore, there exists a surjective homomorphism as k -algebras $\varphi : B \rightarrow A$ such that the following sequence of A -modules is exact and split:

$$0 \rightarrow I \rightarrow B \xrightarrow{\varphi} A \rightarrow 0,$$

where $I := \ker f$. Hence A is finitely generated as a k -algebra. Also by Proposition 1.1 (2), we see that A is a UFD.

We consider a morphism $f : \mathbb{A}_k^n \cong \operatorname{Spec} B \rightarrow X$ defined by a natural inclusion map $\iota : A \rightarrow B$. It follows from Proposition 1.1 (3) that X is nonsingular over k .

Suppose that X is resolvable and $Q(B)$ is separably generated over $Q(A)$. Then f is dominant and generically separable. Hence by Lemma 2.1, we have $\bar{\kappa}(X) \leq \bar{\kappa}(\mathbb{A}_k^n) = -\infty$, which implies that $\bar{\kappa}(X) = -\infty$. $\square$

When we consider the polynomial ring in two variables whose ground field is not necessarily algebraically closed, the following result is useful.

Theorem 2.4. (cf. [4] or [2, Theorem 5.2]) *Let K and k be fields such that K is separably generated over k . Suppose that A is a commutative k -algebra for which $K \otimes_k A \cong K^{[2]}$. Then $A \cong k^{[2]}$.*

The following is the main result in this paper.

Theorem 2.5. *Let k be a field of characteristic zero and let $B := k^{[3]}$ be the polynomial ring in three variables over k . Then every retract of B is isomorphic to the polynomial ring.*

Proof. Let A is a retract of B and let $d := \operatorname{tr.deg}_k(A)$. Clearly, if $d = 0$, then $A = k$. By Proposition 1.1, A is algebraically closed in B . Hence, if $d = 3$, then $A = B = k^{[3]}$. If $d = 1$, then we already know that $A \cong k^{[1]}$ by [1, Theorem 3.5].

Suppose that $d = 2$. Let K be an algebraic closure of k . Set $A_K := A \otimes_k K$ and $B_K := B \otimes_k K$. It follows from Lemma 1.2 that A_K is also retract of $B_K = K^{[3]}$. Set $X = \operatorname{Spec} A_K$. By using Lemma 2.3, we have X is a nonsingular, factorial surface over K and $(A_K)^* = K^*$. Therefore it follows from Theorem 2.2 that $A_K \cong K^{[2]}$. Applying Theorem 2.4 for A_K , we have $A \cong k^{[2]}$. $\square$

Remark 2.6. In Lemma 2.3, we don't know whether $Q(B)$ is separably generated over $Q(A)$ or not in general. Of course, if it is true in general, then Theorem 2.5 holds true for any characteristic.

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